



FUTURE ROBOTICS

ARE YOU READY FOR
THE MACHINE REVOLUTION?

ANNUAL BOOK 25-26

FCR



CREATIVE AND INNOVATION CELL IN ROBOTICS (CICR)

BEYOND IDEAS. BUILDING IMPACT.



A glimpse into a year of curiosity,
creativity and commitment
that moved us forward.



PUBLISHED BY
Creative and Innovation
Cell in Robotics (CICR)



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FACULTY COORDINATORS
Dr. Parul Arora, Dr. Harish Bishwakarma



DESIGN & LAYOUT
CICR PR & Social Team





INNOVATE. INSPIRE. IMPACT.

ANNUAL BOOK 2025-26



Are You Ready for the Machine Revolution?

A publication by the
Creative and Innovation Cell in Robotics (CICR)



From the sparks of curiosity in our labs to the boldest visions of tomorrow, this book captures the journey of every innovator in CICR. Welcome to a world where imagination meets engineering.

– The CICR Team



INNOVATE

Ideas that
inspire change.



CREATE

Building solutions
for a better tomorrow.



IMPACT

Engineering a
smarter future.



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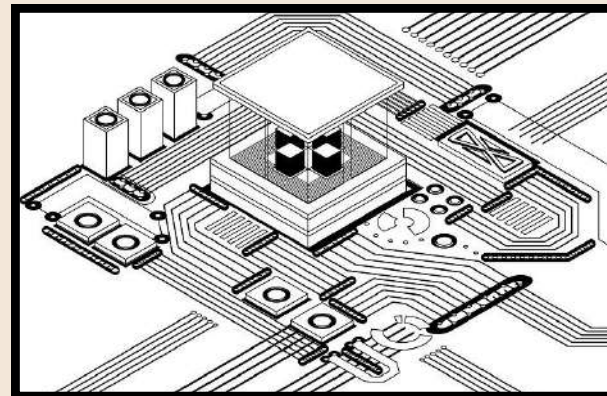
Leadership & Vision

This chapter introduces CICR, its objectives and journey, shares messages from our mentors, and presents our Vision, Mission, and the team leading the club.

01

02 Technical Architecture

This section explores the engineering domains we work in—AI, IoT, and automation—while also showcasing our process of prototyping, digital design, and simulation before fabrication.



Projects Showcase

A showcase of the innovative hardware, robotics, and IoT solutions built by our members this year.

03

04 Events & Activities

A recap of our workshops, community activities, and achievements, including participation in Plinth '25, contributions to Synergy events, skill-building sessions, and the induction of new CICR members.



Administration & Future Vision

An inside look at the club's operations, covering finances, resource management, team growth, challenges, lessons learned, and our vision for 2025–2026.”

05

Message

From **THE DEAN & HOD**



It is a matter of immense pride to observe the steady and commendable progress of the Creativity and Innovation Cell in Robotics (CICR), as a vibrant reflection of the vision and values of the Department of Electronics and Communication Engineering.



Dr. Arti Noor

By effectively integrating theoretical foundations with practical implementation, CICR fosters a culture of innovation, research orientation, and technological excellence. The interdisciplinary initiatives undertaken by the students reflect intellectual rigor, technical competence, and a mindset essential for addressing contemporary engineering challenges.

I appreciate the dedicated efforts of the faculty coordinators and students in nurturing this ecosystem of excellence and extend my best wishes for the continued advancement and distinction of CICR.

Message

FROM THE FACULTY COORDINATOR

We are extremely pleased to present the inaugural issue of CICR hub annual book. This book will reflect the creativity, dedication, and hard work of every student involved in

CICR hub. CICR hub has consistently fostered a culture of innovation and hands-on learning among students. What makes this hub truly special is the enthusiasm of our students who not only conceptualize and develop innovative projects but also confidently showcase them. Their participation in inter-college events, competitions, and exhibitions reflects their technical competence, teamwork, and determination to excel beyond classroom learning.

This book captures the journey of ideas transforming into impactful projects. Our heartfelt congratulations to all the students.



Dr. Parul Arora



Dr. Harish Bishwakarma

The CICR Hub provides students with a practical, hands-on approach alongside their academic studies.

The hub aims to address real-world engineering challenges and develop innovative solutions that are beneficial to society.

By integrating academic learning with practical exposure, it fosters enthusiasm for applied knowledge and strengthens students' ability to translate theoretical concepts into real-life applications.

Message FROM JYC Faculty Coordinators

Watching this club grow over the years has been one of the more rewarding things about being associated with this institution.

CICR has always been more than a technical society. It is a space where students learn to think, fail, rebuild, and ultimately create things they are proud of. The effort that goes into every project, every competition, and every late night in the lab does not go unnoticed.



Dr. Pankaj Shrivastava



Dr. Vinay Anand Tikkiwal

This annual book is a record of that effort. We hope it serves as a reminder to every member of what they are capable of, and as an inspiration to those who are just starting out.

Our best wishes to the entire CICR family. Keep building, keep growing.

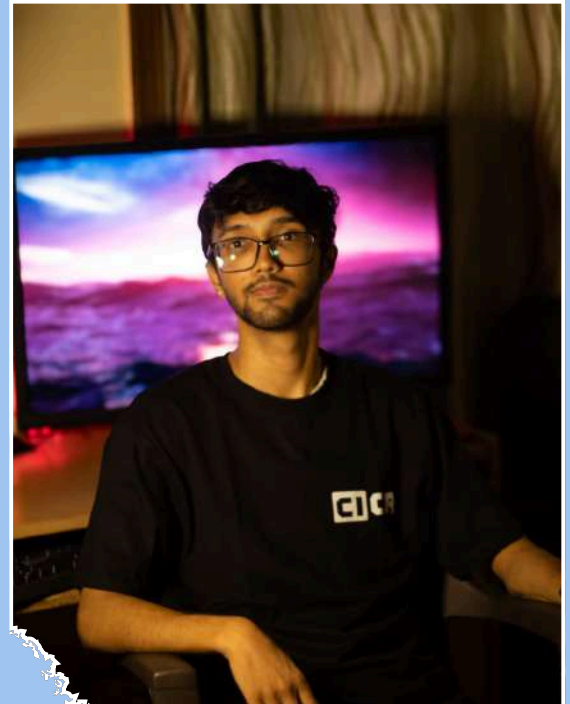
Message From **THE ADVISOR**



“

Hello readers,
Welcome to the first edition of the CICR Annual Book (2025-26) ,a reflection of our journey, growth, and the work that defines who we are.

CICR has been an integral part of my life for the past four years. When I first joined college, I was a beginner making basic mistakes and still figuring things out. But everything started to change when I came across this society. The environment, the people, and the shared drive to build and learn made a real difference. At that time, the society wasn't what it is today, but it had something powerful people who were genuinely eager to grow. I joined, started working on projects, and over time, gained not just technical skills but confidence and direction. One of the biggest strengths of CICR is that it becomes more than just a club it turns into a strong support system. A place where you build, experiment, fail, improve, and grow together. Over the last four years, CICR has grown tremendously both in scale and impact. Today, we are working across multiple domains including robotics, drones, web development, machine learning, artificial intelligence, IoT, 3D printing, embedded systems, and more.



Archit Jain

With growing support from the college, better resources, and an expanding inventory of components and tools, the scope of what we can build is only increasing. We are also open to collaborations and always ready to connect with people who share the same passion for innovation. I truly believe this society has the potential to reach much greater heights and evolve into something even bigger in the future.

This book captures a glimpse of our journey from projects and events to achievements and experiences. I encourage you to explore it, go through the links, and see what CICR is all about.

Thank you.”

THE STORY SO FAR

Who We Are

The Creative and Innovation Cell in Robotics (CICR) at the Jaypee Institute of Information Technology is an organization based on curiosity and innovation.

Our organization unites enthusiastic students interested in Robotics, Internet of Things (IoT), Artificial Intelligence/Machine Learning (AI/ML), Development, and Public Relations to create rather than merely study technological knowledge.

Not merely a club, the CICR is an arena of idea development into practical systems.





THE YEAR 2025-26

With the help of flagship robotics contests, technical seminars, multi-college involvement, and basic skills training, CICR was able to increase its footprint in the realms of Robotics, Electronics, Computer-Aided Design (CAD), and technology applications, emphasizing implementation, systems thinking, and teamwork.

A Culture of Collaboration



Juniors were mentored by seniors in areas such as hardware design, PCB design, CAD modeling, Web-Development, AIML, and robotics system integration. The teams cooperated in creating competition arenas and constructing wireless robots.

This approach blended discipline with creativity, allowing all team members to have their say in technical achievements.



Looking Forward

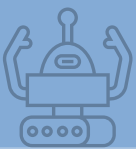
CICR, with its aim of building a link between theory and practice, goes on making a difference by molding the future technocrats, thinkers, and decision-makers.

"We don't just learn technology - we create with it."

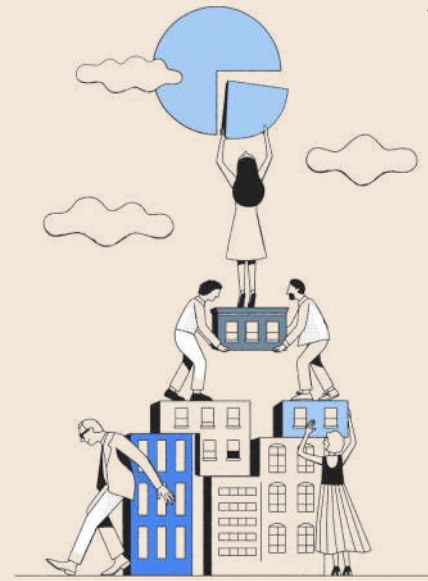
VISION

CICR is envisioned as a recognized Center of Excellence in robotics and automation whose graduates are not only experts in designing and implementing robots but also understand the implications of their creations within society and the environment.

It will be a major influence in technology that will determine the future of our university, our country, and others far beyond it.



MISSION



1

To create a collaborative workspace equipped with advanced tools and resources for designing, building, and testing robotic systems.

2

To encourage creativity and originality in every stage of the engineering process, from ideation to deployment.

3

To build solutions that address real-world challenges in industries such as healthcare, manufacturing, defense, and environmental protection.

4

To prepare students for national and international competitions, ensuring they uphold professionalism and technical excellence.

5

To develop members technical expertise in electronics, programming, mechanics, and artificial intelligence.

The CICR STORY

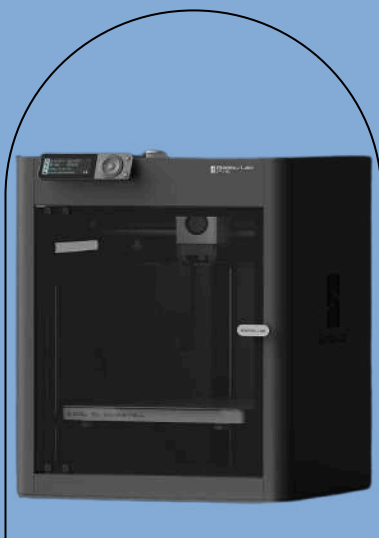
DESIGN

DEVELOP

DOMINATE

LATEST

Installation of Bambu lab P1S 3D Printer



August 2025 saw an expansion in the lab's capacity to prototype and fabricate through the acquisition of a Bambu Lab P1S 3D printer. The inauguration ceremony was conducted by the Faculty Coordinator, Dr. Arti Noor, representing a significant milestone towards enhancing facilities for hands-on innovation and prototyping. Through the inclusion of a Bambu Lab P1S 3D printer, students can make their digital models tangible.



As a Community

Fostering Creativity

CICR stands for Creative and Innovation Cell in Robotics, which is a student-oriented environment that facilitates learning through making and tinkering. Some of the domains it operates in include Robotics, IoT, AI/ML, Web Development, and Computer-Aided Design.

With all these facilities at hand, it was possible for the students to convert their theoretical knowledge into practical applications.

Where Innovation Meets Purpose

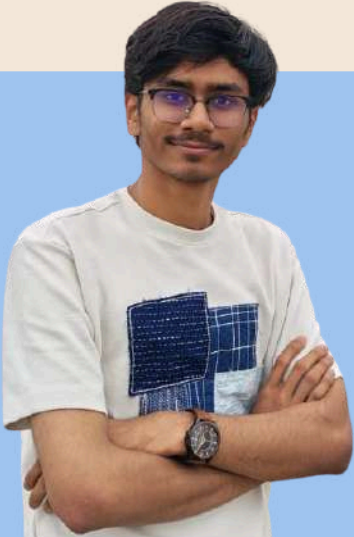
At CICR, the goal is not only to assist students to become more technical but also to prepare them for future tasks. Through consistent construction of projects, mentorship, and the chance to showcase their institution, students acquire an innovative and collaborative way of thinking.





THE LEADS

Meet Our Team



Yasharth Singh

Coordinator



Taarini Mishra

PR Head



Hardik Gupta

Technical Head



Aman Raj

Technical Head



Vaibhav Katariya

Management Head

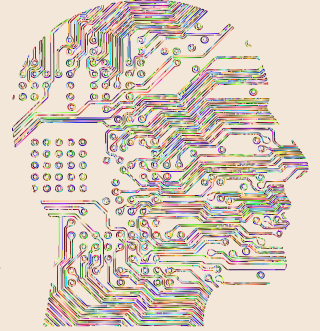


Aryan Gupta

Management Head

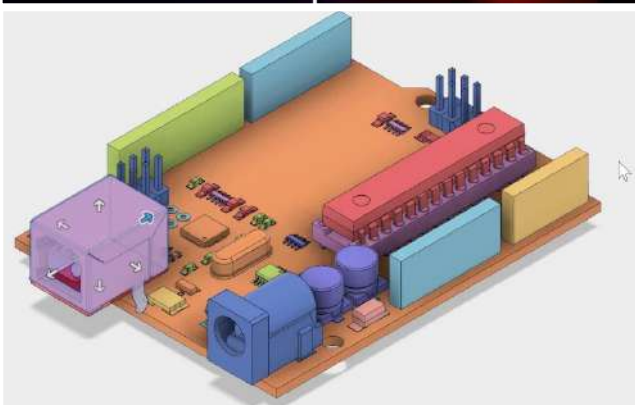
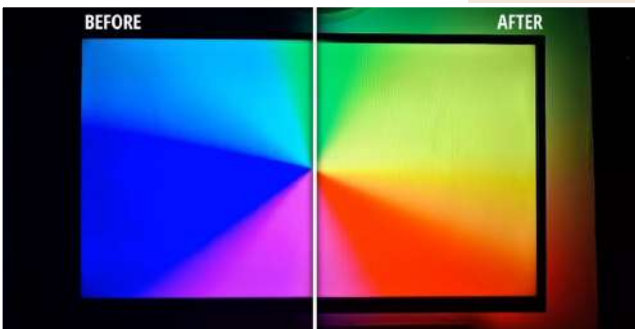


This isn't just a team it's a system.
Each lead owns a domain, drives innovation, and ensures that ideas don't stay ideas.
From code to communication, everything here is engineered for impact.



Domains *We Explore*

Every contribution, no matter the size, helps us continue our vital work.
Here's a snapshot of how your kindness has made a difference this month:



Integrating
Artificial
Intelligence for
Intelligent
Robotics and
Automation
AI

Ft. SKULLY, SAHYATRI

Designing Robust
Mechanical
Systems

**HARDWARE &
AUTOMATION**

Ft. DEQUE

IoT for
Monitoring and
Remote Control

IOT

Ft. Ambient Light

Designing and
Prototyping
Efficient Electronic
Systems

**ELECTRONICS &
CIRCUIT DESIGN**

Ft. ESP-32 Mini Drone

Data-Driven
Models &
Predictive
Systems

ML

Ft. JARVIS

Enabling
Machines to See
& Interpret the
World

**COMPUTER
VISION**

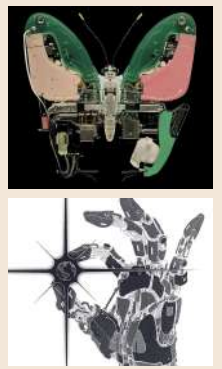
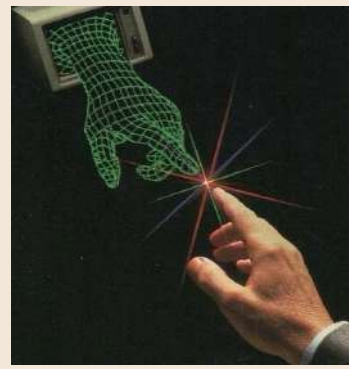
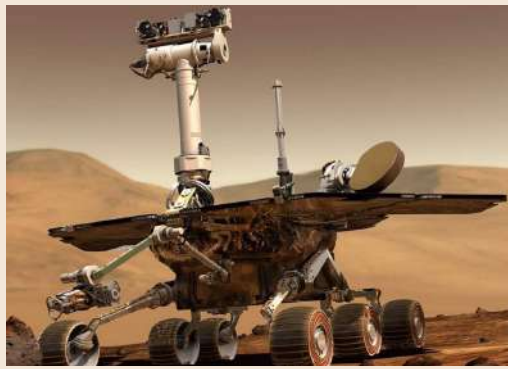
Ft. JARVIS

Building Scalable
Software Solutions
for Real-World
Applications

SOFTWARE

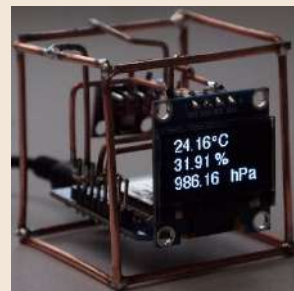
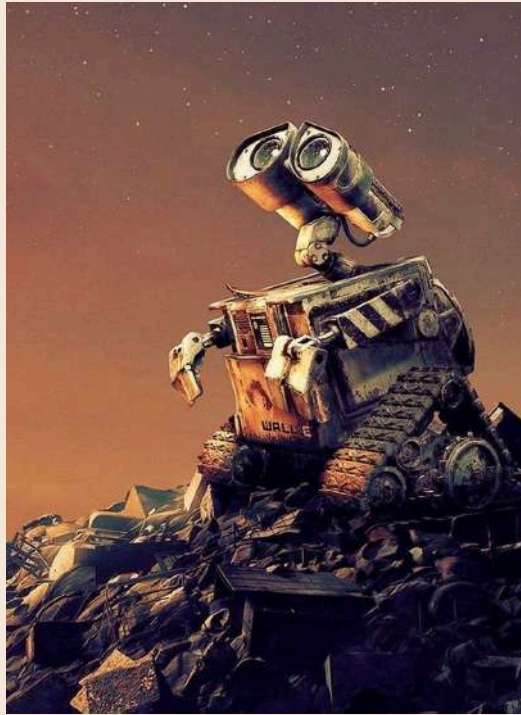
Ft. CICR Connect





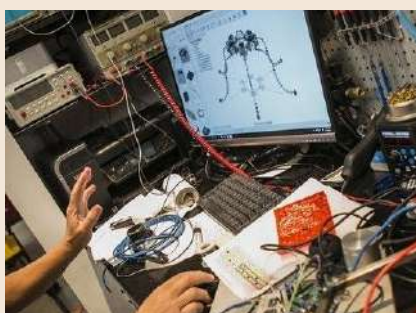
BEHIND THE BUILD

From idea → prototype → real system
We don't just build projects—we engineer them through a structured and iterative process. Every idea begins as a concept, evolves through design and experimentation, and is refined through continuous testing until it becomes a reliable, real-world solution. Our focus is not just on building, but on understanding, improving, and delivering systems that actually work.



Rapid Prototyping & Fabrication

We transform concepts into physical systems using 3D printing, electronics, and mechanical assembly. This stage emphasizes speed and experimentation—allowing us to test ideas early, identify flaws, and iterate quickly. By building fast and refining continuously, we ensure that designs are both practical and efficient.



Digital Design & Simulation

Before fabrication, we design and analyze every component using CAD tools and simulation platforms. We model systems, test performance under different conditions, and optimize structures for accuracy and reliability. This reduces errors, saves resources, and ensures a smoother transition from design to reality.

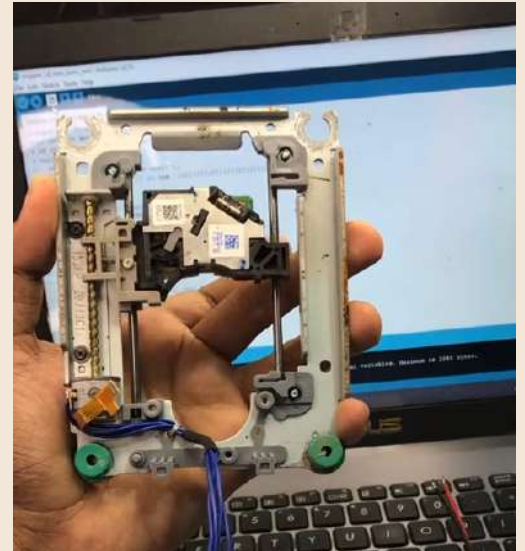


AI & Intelligent Systems

We build systems that go beyond predefined instructions. These systems can learn, adapt, and evolve with experience. By combining artificial intelligence with robotics, we explore how machines can perceive their surroundings, understand patterns, and make informed decisions in real time. Our work is driven by curiosity about how we can make machines not just execute tasks, but also understand them. From intelligent automation to decision-making systems, we focus on creating solutions that become smarter over time and perform reliably in real-world scenarios. For us, AI isn't just about complexity. It's about making intelligence practical, usable, and impactful.

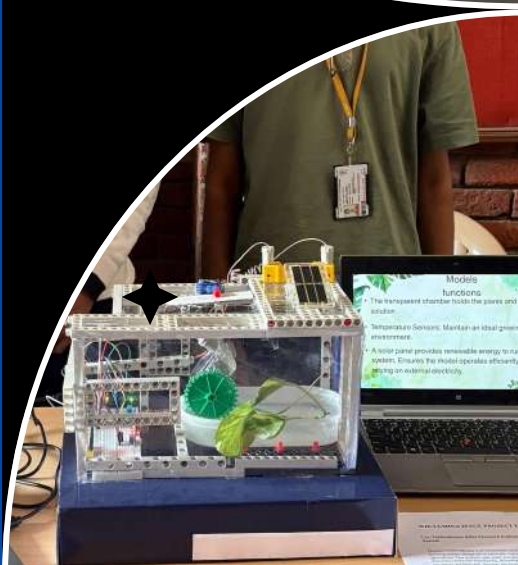
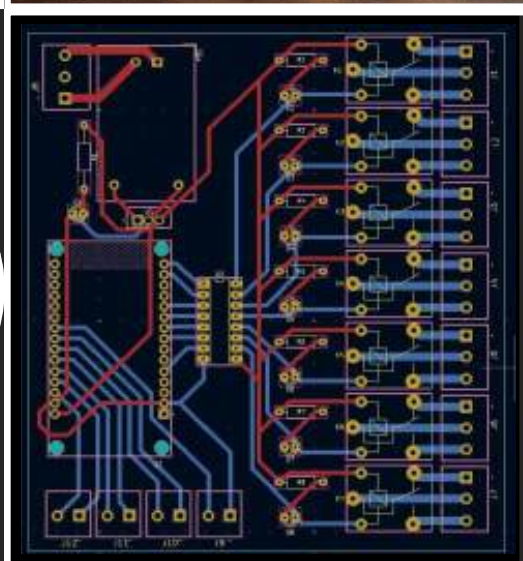
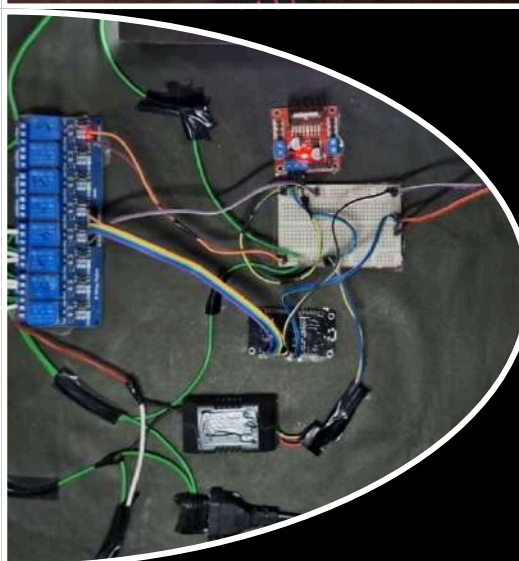
Embedded Systems & IoT Integration

This is where our ideas move from theory to the real world. We design and build embedded systems that connect sensors, actuators, and processors into a unified unit that can interact in real time. Using platforms like microcontrollers and IoT modules, we create systems that monitor environments, share data, and respond quickly to changing conditions. Every project here focuses on making things work beyond the screen, whether it involves automation, remote control, or smart sensing. It's the layer where engineering becomes real, where circuits, code, and logic combine to create something you can see and interact with.



Software & System Development

Behind every working system is a layer of software that keeps everything together. We design and develop the logic that connects hardware, processes data, and creates meaningful user interactions. From control systems to user interfaces, we focus on building software that is stable, efficient, and easy to use. In the end, even the best hardware needs thoughtful software to truly come alive.

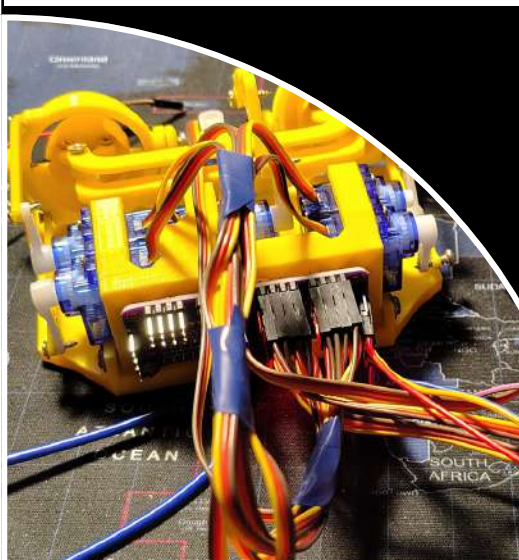


 **LABOURLINK**
DIAL. HANGUP. GET HIRED

PROJECTS

2025--2026

 **Sahyatri**



CI CR

SKULLY

ANIMATRONIC ROBO

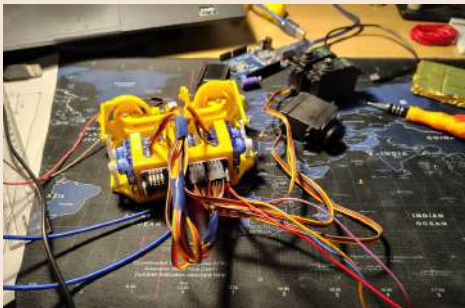
This project implements a **vision-enabled** animatronic robotic head built on a modular, servo-actuated mechanical framework designed for expressive human-robot interaction. The system integrates **real-time perception**, **motion control**, and **speech synthesis** to produce synchronized behavioral responses.

At the hardware level, the platform is constructed around a **custom red 3D-printed skull chassis**, engineered to support modular component mounting and iterative mechanical refinement. The actuation system consists of **10 high-precision servo motors**, each assigned to specific degrees of freedom including jaw articulation, head orientation, and auxiliary expressive movements. This distributed actuation architecture enables smooth, coordinated motion with fine positional control.

The system architecture has evolved from a microcontroller-centric design using **Espressif Systems's ESP32** to a heterogeneous compute model centered on the **NVIDIA Jetson Orin Nano**, which now serves as the primary processing unit. This transition significantly expanded computational capability, enabling onboard execution of computer vision models, sensor fusion, and higher-level behavioral logic.

For perception, the platform initially utilized the **Adafruit Memento Camera Board**, but has since been upgraded to an **Intel RealSense Depth Camera**, providing stereo depth sensing and spatial awareness.

*Selected among the **Top 10** in the **CircuitDigest Smart Home and Wearables Contest 2025** with the project "Skully AI-Driven Assistant." Animatronic Human-like Home*



Project Contributors

ARCHIT JAIN
BUSHRA KHANAM
SHREYASH SINGH

TECH STACK

Hardware: ESP32, Servo Motors, 3D Printed Mechanical Frame, Jetson Nano, Intel Realsense

Software: Python, Embedded C, ROS2, CV

Communication: Wi-Fi, ESP-NOW

Tools: Fusion 360, 3D Printing, Hardware Prototyping



SAHYATRI SMART OCCUPANCY MONITORING SYSTEM

SahYatri is an Edge AI powered IoT solution developed for **real time bus fleet management** and occupancy monitoring in Indian public transportation

1

REAL-TIME OCCUPANCY DETECTION

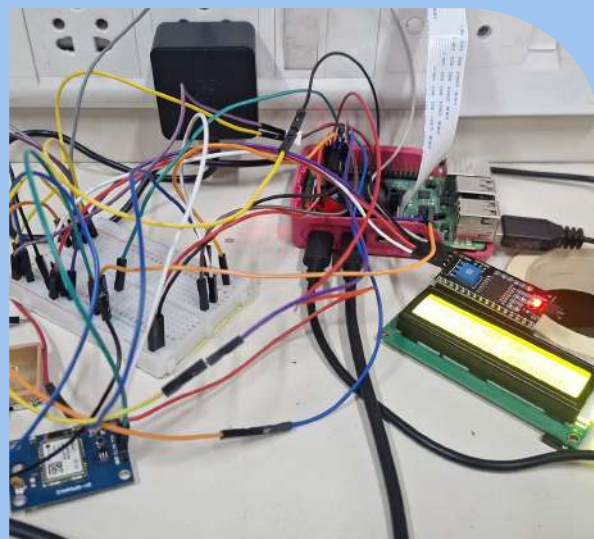
The system performs on-device passenger counting using **Edge AI**, enabling sub-second detection without relying on continuous internet connectivity. Video processing runs locally on the device, ensuring fast response times and improved privacy while maintaining reliable occupancy monitoring during transit operations.

2

EDGE AI & SENSOR MONITORING

In addition to vision-based detection, SahYatri integrates sensors such as GPS, IMU, and temperature modules to provide **fleet location tracking** and **operational insights**. Data is buffered offline and automatically synchronized when connectivity becomes available, ensuring uninterrupted monitoring even in low network areas.

*SahYatri secured **1st place at BITBOX 5.0**, organized by **Google Developers Group - JIIT128**, winning **₹20,000**, and was also selected for the **Project Exhibition at IIT Roorkee** during Cognizance'26.*



PROJECT TECH STACK SahYatri

Hardware

- Raspberry Pi 4
- Pi Camera V2
- GPS Module
- IMU Sensor
- Temperature Sensor

Backend

- Node.js Server
- PostgreSQL Database

Cloud Sync

- Firebase

Edge AI

- YOLOv5 Passenger Detection Model
- Python + OpenCV Image Processing

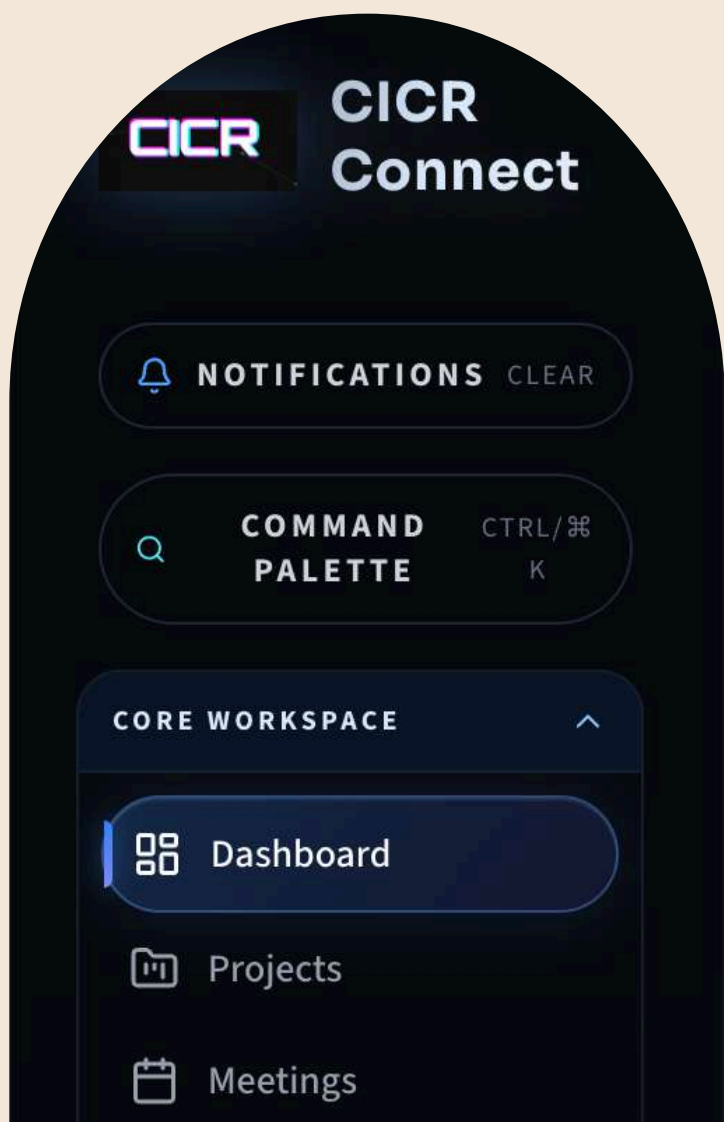
Interfaces

- React.js Web Dashboard
- Flutter Mobile Application

Project Contributors

YASHARTH SINGH
RAKSHIT SUNEJA
AMAN VERMA
SHIVAM KASHYAP





CICR CONNECT UNIFIED ECOSYSTEM PLATFORM

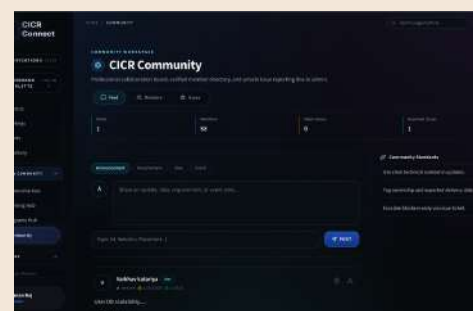
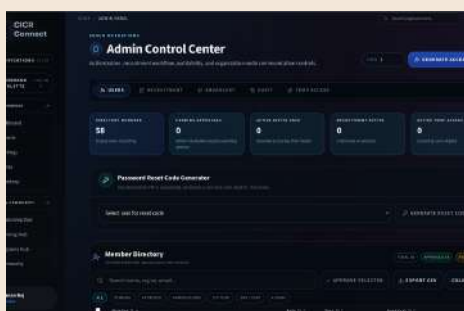
CICR Connect is a comprehensive **web** and **mobile** platform developed for secure **ecosystem management**, replacing scattered workflows with centralized, role-based operations.

1 UNIFIED OPERATIONS & IDENTITY

Built on the MERN stack and Expo, the platform **centralizes** Projects, Events, and **hierarchy-based** task management. It secures user access with **invite-based authentication** and features a QR-based DigiCard for seamless identity sharing among **students, alumni**, and mentors.

2 GOVERNANCE & SCALABLE SECURITY

Admins maintain **strict ecosystem** control through moderation tools, audit logs, and access workflows. Designed for reliability, the system supports user growth via **Learning Tracks** and **mentorships**, backed by robust security measures including data validation and rate limiting.



Project Contributors

AMAN RAJ

TECH STACK

Frontend : React 19, Vite 7, React Router 7, Tailwind CSS 4, Framer Motion, Axios

Backend: Node.js 22, Express 5, MongoDB, Mongoose, JWT, bcryptjs, express-validator, Nodemailer

AI: Custom LLM fine-tuned on proprietary CICR data.

Tooling: ESLint 9, Node test runner



J.A.R.V.I.S - THE INTERACTIVE AI FACE

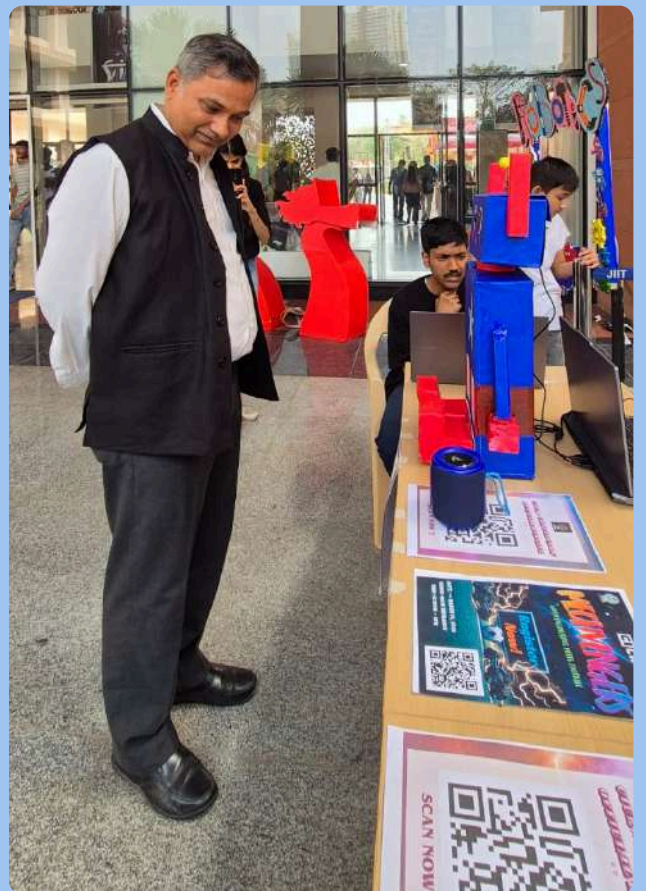
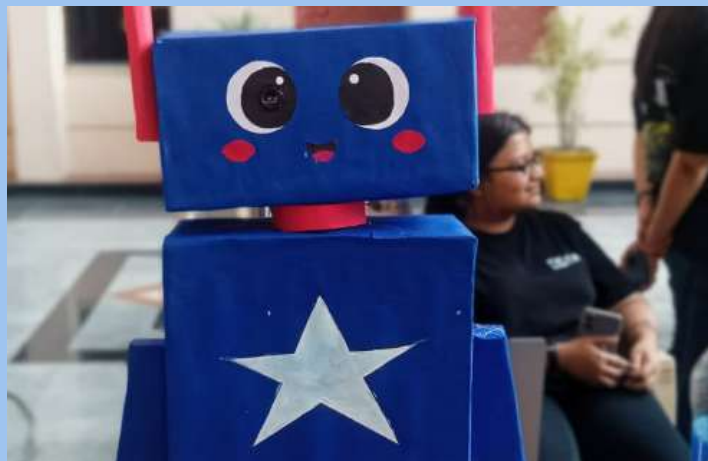
At Converge 2026, the CICR stall showcased J.A.R.V.I.S., a **real-time interactive AI host** that personalized the fest experience. It **greeted attendees by name**, engaged them with **witty conversations**, and guided them through Marvel-themed events like Spider Sense, Thor's Trail, and Stark Speedway.

1

EXCELLENCE AND ACHIEVEMENT

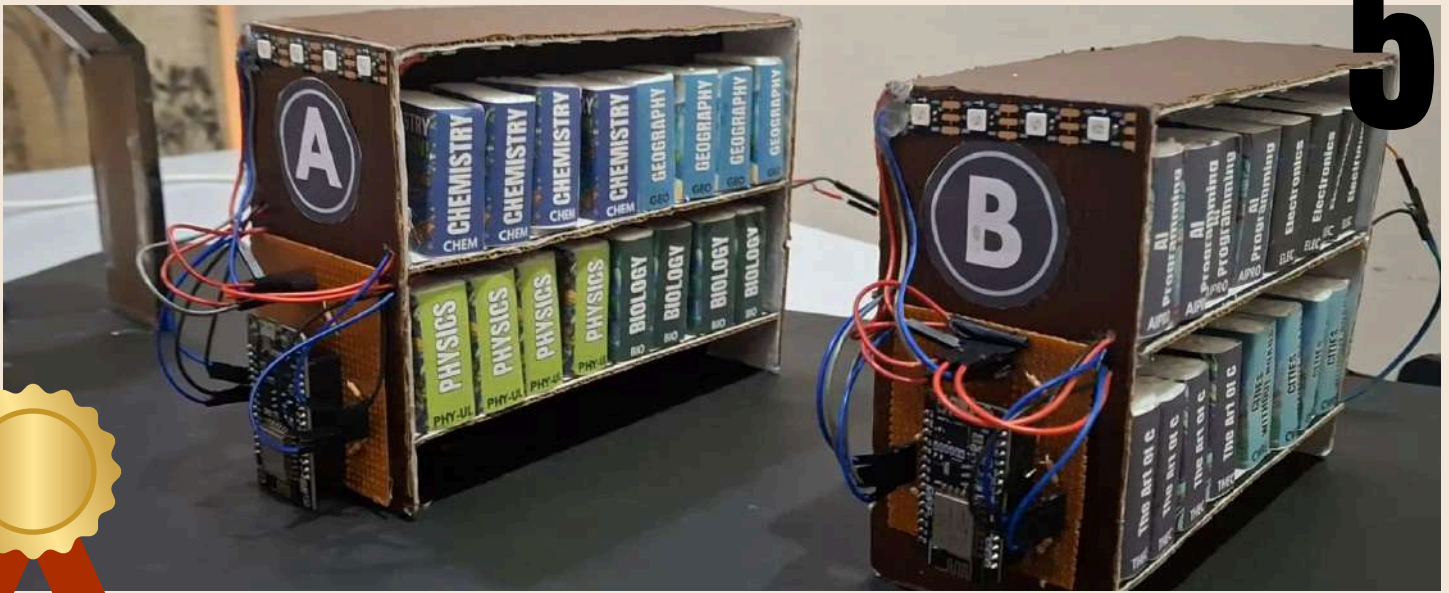
Powered by a **low-latency, multi-threaded system**, J.A.R.V.I.S. combined DeepFace (FaceNet512) with a FAISS database for instant **facial recognition**, and a SQLite database for user profiles. Voice interaction was enabled via Silero VAD, **Sarvam AI's Speech-to-Text and Text-to-Speech**, and a Groq-powered LLM for fast, context-aware responses.

The system delivered seamless, real-time performance, demonstrating how advanced AI pipelines can operate efficiently in dynamic, high-energy environments.



PROJECT CONTRIBUTORS
ARYAN VARSHNEY





FIND MY BOOK

AI & IoT SMART LIBRARY NAVIGATION SYSTEM

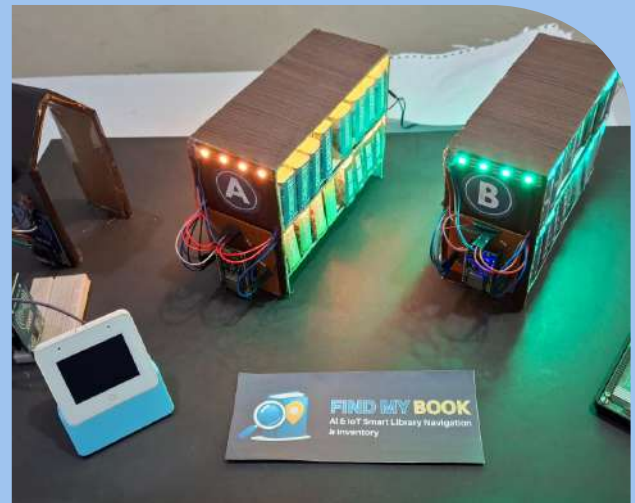
Students often struggle to locate books in large library racks. While digital catalogs show availability, they rarely show the exact location. Find My Book is a smart system that improves library operations by providing real-time book location guidance, automated issuing/returning, and security feature.

1

EXCELLENCE AND ACHIEVEMENT

Aman Gandotra secured 2nd Position and INR 70,000 in the Smart Home and Wearables Contest 2025 with his project Find My Book - AI & IoT Smart Library Navigation System. The system uses ESP32-S3 Box 3, ESP8266 modules, RFID technology, and a Raspberry Pi backend to help users locate books quickly. By combining IoT shelf navigation, RFID issuing/returning, and an anti-theft mechanism, the project makes library management smarter and more efficient.

*Find My Book secured **2nd position** in the **CircuitDigest** Smart Home and Wearables Contest **2025** and **won INR 70000**.*



PROJECT TECH STACK FIND MY BOOK

Hardware

- ESP32-S3 Box 3
- ESP8266
- Raspberry Pi

Backend

- Flask Server
- SQLite Database
- REST APIs

Interfaces

- Touchscreen Kiosk
- Web Dashboard
- QR Mobile Interface

Technology

- RFID
- IoT

Edge AI

- Python
- Embedded C
- LVGL UI Library
- SquareLine Studio UI

Project Contributors
AMAN GANDOTRA





LabourLink

Dail.HangUp.GetHired

LabourLink is a technology-driven platform designed to **bridge the gap** between labour **contractors** and **daily-wage workers**. It enables workers to register their availability through a simple **missed call**, eliminating the need for **smartphones** or **internet access**. Contractors can efficiently hire workers through a **centralized web dashboard** based on skills, availability, and location.

1

THE IMPACT

Workers receive **automated voice calls** and **SMS** notifications with job details such as location, reporting time, and wages. The platform maintains worker **records**, **availability tracking**, and hiring **history**, ensuring **transparency** and trust. By combining accessibility with **automation**, LabourLink simplifies hiring in the unorganized sector and **reduces delays in workforce coordination**.

*LabourLink won the **Best Fresher Award** at **BitBox 5.0 (GDG-JIIT)** and received a **cash prize of ₹5000**, recognizing its real-world impact and innovation in solving labour accessibility challenges.*



PROJECT TECH STACK LabourLink

Frontend

- React
- Framer Motion
- TailwindCSS

Backend & Database

- Flask
- MongoDB

Interfaces

- Web application
- NGrok

Project Contributors

ASHUTOSH VISHWAKARMA
DHRUVI GUPTA
NIDHI CHAUHAN





DEQUE

SMART RFID PAYMENT SYSTEM

Deque enables fast, contactless payments using RFID tags attached to products. Users can scan and pay seamlessly without traditional billing queues, making transactions quick and efficient.

1

REAL-TIME THEFT DETECTION

Each product's RFID is linked to its payment status. At exit points, the system verifies whether items have been paid for. If an unpaid item is detected, it instantly triggers alerts, helping prevent theft automatically.

2

SECURE & SCALABLE SYSTEM

Each product's RFID is linked to its payment status. At exit points, the system verifies whether items have been paid for. If an unpaid item is detected, it instantly triggers alerts, helping prevent theft automatically.

DEQUE secured **3rd place** at **HACKIN'WINTER**, organized by **DSC - IIIT62**, winning **₹2,000**.



PROJECT TECH STACK DEQUE

Hardware

- Raspberry Pi 4
- Pi Camera V2
- RFID Module

Backend

- Node.js Server
- MongoDB
- Flask

Edge AI

- YOLOv5 Passenger Detection Model
- Python + OpenCV Image Processing

Interfaces

- Web application
- NGrok

Project Contributors

VANSHIK JAIN
SOMYA JAIN
MOHD KASHIF KHAN
AARYA GARG

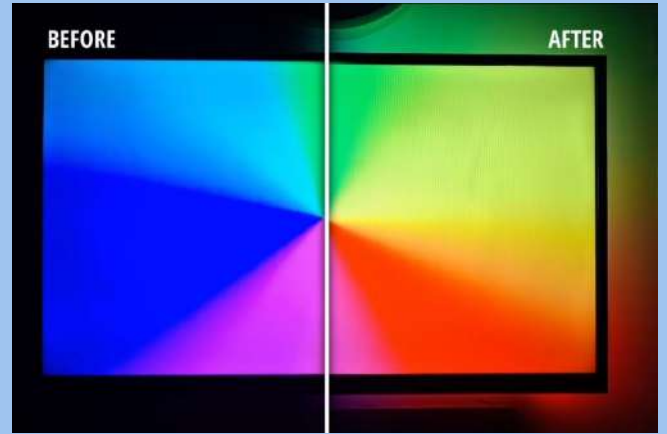


DIY Ambient Light TV

During this project, a standard television was transformed into an Ambient Light TV, showcasing how dynamic lighting can enhance the viewing experience by extending on-screen colors into the surrounding environment.

The system was built from scratch using a Raspberry Pi, WS2812B addressable LED strips, an HDMI splitter, and a video capture card. The HDMI signal from the set-top box was split, allowing the Raspberry Pi to capture video frames in real time and analyze the colors at the edges of the screen.

Using Hyperion, the Raspberry Pi processed these frames and converted the color data into an array of signals. These signals were then sent to the LED strips, which reproduced the corresponding colors behind the television. Operating at approximately 60 frames per second, the system maintained smooth performance with minimal delay, ensuring that the ambient lighting stayed perfectly synchronized with the on-screen visuals. This resulted in a seamless and immersive effect, where the content on the display blended naturally with the surrounding light.



Project Contributor

AMAN GANDOTRA



8



Project Contributor

ARCHIT JAIN
VAIBHAV KATARIYA
YASHARTH SINGH

PAYLOAD DELIVERY DRONE

This project presents a payload delivery drone developed to transport small items using a stable quadcopter platform. The drone is built on an F450 frame and powered by DJI 920Kv motors paired with ESCs, ensuring reliable flight performance and control. A Pixhawk flight controller integrated with a GPS and compass module enables precise navigation and supports autonomous flight capabilities. The system also incorporates a servo-based gripper mechanism for controlled pickup and release of payloads, making it suitable for small-scale delivery and experimental applications.

The design focuses on modularity and ease of maintenance, allowing components to be easily upgraded or replaced. Basic fail-safe features such as return-to-home and stable hovering improve operational safety, while the lightweight structure enhances efficiency and flight duration. This project demonstrates practical implementation of drone-based delivery concepts and provides a foundation for further exploration in aerial automation systems.

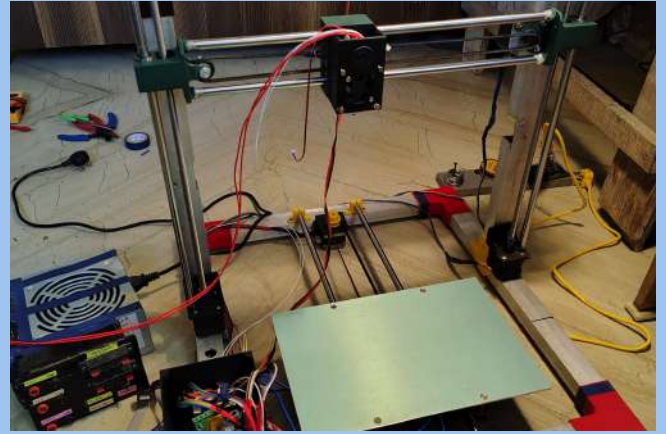
9

CARTESIAN 3D PRINTER

This project is a custom-built 3D printer developed using a LEGO-like modular assembly system that simplifies construction, modification, and maintenance. The entire mechanical structure is fully designed in CAD, ensuring accurate alignment, rigidity, and efficient motion planning. The system is powered by an Arduino Mega 2560 combined with a RAMPS 1.4 control board, providing a reliable platform for motion control and firmware integration.

Multiple stepper motors drive the X, Y, and Z axes along with the extrusion mechanism, enabling precise positioning and consistent printing performance. The modular architecture allows individual parts to be easily assembled, replaced, or upgraded, making the printer highly adaptable for experimentation and future enhancements.

Additionally, it uses open-source firmware for compatibility and easy configuration, with optimized design for reliability and learning.

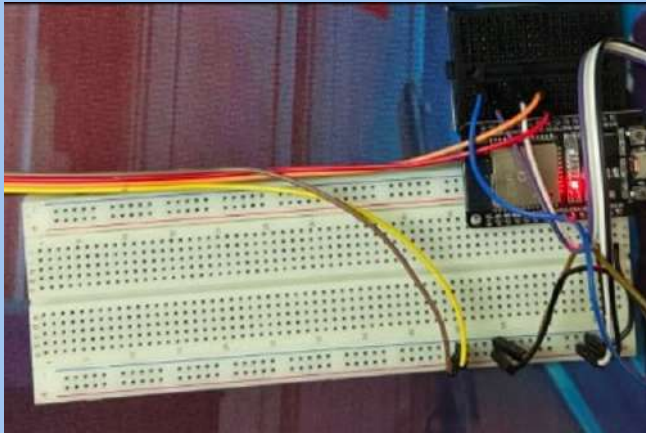


Project Contributor

ARCHIT JAIN



10



Project Contributor

VAIBHAV KATARIYA
ADARSH SHARMA



Smart Automated Water Storage Monitoring and Control System

TANKMIND is an IoT-based smart water tank monitoring and control system that automates water management while improving safety, efficiency, and water quality awareness. It prevents overflow, motor dry runs, pump failures, and water wastage using multiple sensors, automated control, and cloud connectivity for real-time monitoring. Built on an ESP32 microcontroller, it uses a waterproof ultrasonic sensor for water level detection, a YFS201 flow sensor to measure actual flow, and TDS and pH sensors to monitor water quality, with all data accessible through a mobile interface.

The system also automates tank and motor operations by turning the motor on when water levels are low and switching it off when the tank is full. It detects issues like dry runs and blockages, monitors water quality, and sends alerts for unsafe conditions. A manual override provides emergency control, while the overall automation and monitoring help reduce water and energy wastage.

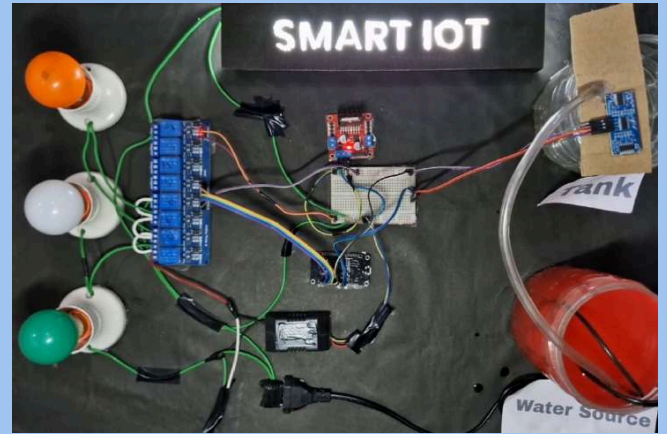
11

Smart IoT Home Automation & Water Management System

This project focused on developing a smart IoT-based system using the ESP8266 to enable real-time control of home appliances and efficient water management through a personalized mobile web dashboard. The system allows users to remotely control 220V lighting over Wi-Fi using relay modules, offering convenient and energy-efficient operation.

In addition to lighting control, the project includes an intelligent water tank management system built using an HC-SR04 ultrasonic sensor, a water pump, and a dual-tank setup. The system operates in both manual and automatic modes. In manual mode, users can directly control the pump, while in automatic mode, the system monitors water levels and activates or deactivates the pump based on predefined thresholds—preventing overflow and protecting against dry running.

Real-time water level data is continuously measured and displayed on the dashboard in centimeters, allowing users to monitor the system remotely with ease. Overall, this project demonstrates the practical integration of IoT, embedded systems, and automation to improve everyday efficiency, convenience, and resource management.



Project Contributor

AMAN GANDOTRA



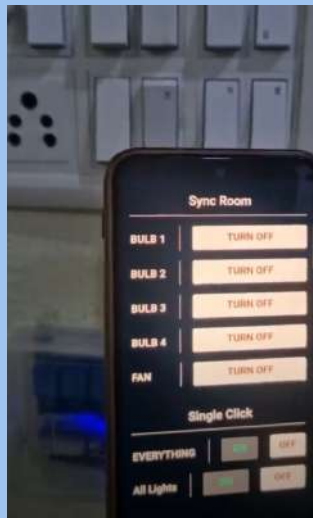
12



IOT-BASED SMART SWITCH SYSTEM

This project focused on designing and installing a custom IoT-based smart switch system to automate room lighting and appliances. The system allows devices to be controlled both manually through physical switches and remotely via a self-developed mobile application. Whether at home or away, appliances can be managed conveniently with a single tap. The hardware setup included an 8-relay module for controlling multiple appliances, an ESP8266 Wi-Fi module for wireless connectivity, and AC to DC converters to provide regulated power to the system. A 12V DC converter was used for stable supply, while a recycled 5V converter sourced from an old phone charger demonstrated resourcefulness and cost-effective implementation.

This hands-on project combined hardware integration, circuit design, embedded programming, and practical problem-solving. It reflects the application of IoT principles in real-life home automation and highlights innovation through efficient and sustainable design.



Project Contributor

AMAN GANDOTRA

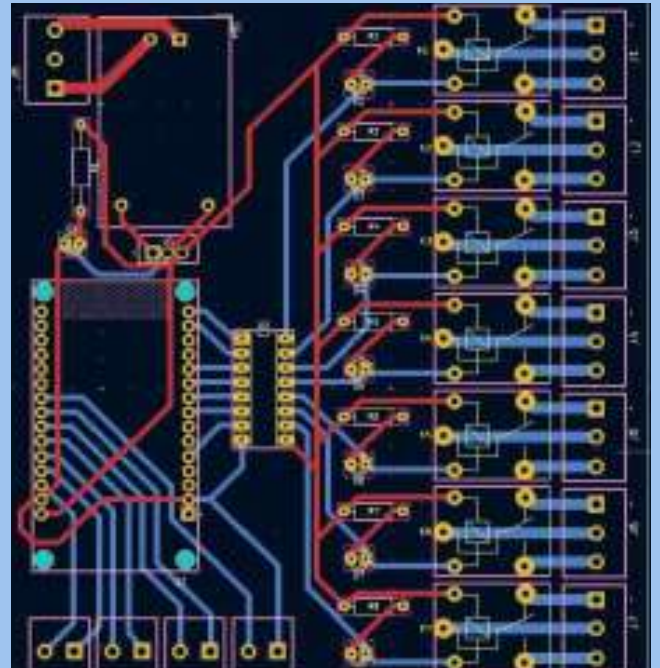


13

MULTI-MODAL AI VOICE ASSISTANT

The Multi-Modal AI Voice Assistant is an open-source, affordable, and privacy-focused smart assistant designed to combine advanced AI-powered voice interaction with IoT-based hardware automation. Developed using Python for software processing and ESP32 for hardware control, the system enables users to manage both computer-based tasks and real-world home appliances through simple and natural voice commands. The assistant can perform functions such as controlling lights or devices, executing system operations, retrieving real-time information, and responding conversationally, creating a seamless interaction between the user, software, and physical devices.

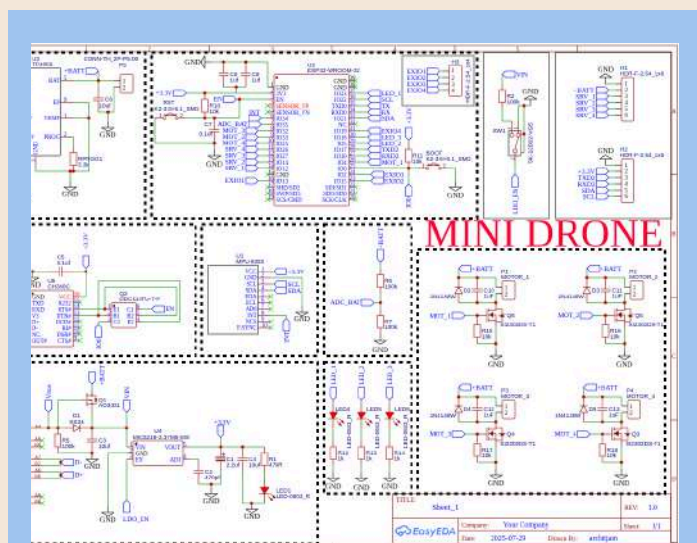
Its modular architecture makes it easy to add new IoT devices, improve voice recognition, or incorporate additional AI features. With its low-cost hardware requirements and scalable design, the project serves as a powerful alternative to commercial assistants while continuing to evolve with future upgrades, improved intelligence, and expanded automation capabilities.



Project Contributor

ARCHIT JAIN

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Project Contributor

ARCHIT JAIN

ESP32 MINIDRONE

The ESP32-Based Mini PCB Drone is a compact quadcopter built around a custom-designed printed circuit board that integrates the drone's core electronics and structural layout. Powered by an ESP32 microcontroller, the system provides sufficient processing capability for flight control while also offering built-in wireless connectivity for future features such as telemetry or wireless control. The drone uses lightweight coreless DC motors, making it suitable for micro-drone applications where size, weight, and efficiency are critical. For flight stabilization, the drone incorporates an MPU6050 inertial measurement unit (IMU), which combines a 3-axis gyroscope and a 3-axis accelerometer to continuously measure motion and orientation. The ESP32 processes this sensor data in real time to adjust motor speeds and maintain stable flight. By implementing sensor-based feedback and control algorithms, the system ensures balanced and responsive drone movement during operation.

A key feature of the project is the custom PCB design, which integrates motor control circuits, power management, sensor interfaces, and microcontroller connections into a single board, reducing wiring complexity and overall weight. This project demonstrates the integration of embedded systems, sensor-based stabilization, and compact PCB design to create a lightweight and efficient drone platform.

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CHIKITSAARM

Gesture Driven Robotic Arm Simulation in VR

This project uses an ESP32 microcontroller connected to an MPU6050 IMU via I²C to capture real-time accelerometer and gyroscope data for motion tracking. The firmware is developed in Arduino IDE using C/C++. A complementary filter fuses accelerometer and gyroscope data to compute stable pitch, roll, and yaw values while reducing noise and drift.

The ESP32 connects to Wi-Fi and transmits orientation data as compact JSON packets over UDP for low-latency real-time streaming. It also hosts an embedded HTTP web server that serves an HTML, CSS, and JavaScript dashboard to visualize IMU orientation and flex sensor readings directly in a browser.

On the receiver side, Unreal Engine processes the incoming UDP JSON packets using custom Blueprint and C++ logic, mapping the orientation data to a 3D robotic arm or hand model in a VR environment to replicate real-world hand movements. The system is designed mainly for educational applications such as medical training, allowing instructors to demonstrate procedures in VR while students observe through immersive visualization.



Project Contributors

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16



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17

CARENEST

CareNest v0.1.0 is an AI-powered maternal and child healthcare platform developed to support pregnant women, newborns, and malnourished children, particularly in rural and underserved communities where access to reliable healthcare is limited. The platform focuses on delivering accessible, culturally sensitive, and voice-enabled assistance, ensuring that even users with low literacy levels can benefit from its services. By combining real-time guidance with localized context, CareNest aims to bridge gaps in awareness, monitoring, and timely medical support.

It offers a wide range of features, including AI-personalized diet planning tailored to the nutritional needs of mothers and children, along with continuous health and growth tracking to monitor development over time. The system also supports vaccine scheduling and nutrition monitoring to ensure that critical milestones are not missed. Users can access hospital locator services to find nearby healthcare facilities, while a multilingual chatbot provides instant assistance in multiple regional languages.

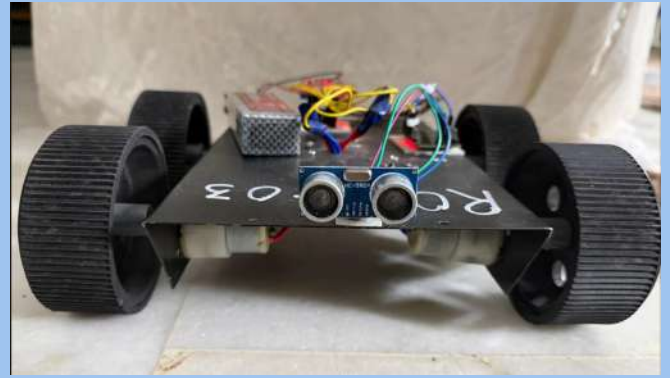
In addition, CareNest includes symptom analysis tools and basic emergency triage guidance to help users identify potential health risks early and take appropriate action.

GESTURE CONTROLLED CAR

We aim to design a gesture-controlled robotic car that operates through simple hand movements, eliminating the need for traditional controllers. A lightweight glove with a gyroscope will detect hand tilt—forward for movement, backward for reverse, and left/right for steering—providing an intuitive user interface.

The glove's motion data will be processed by a microcontroller and transmitted wirelessly (via RF or Bluetooth) to the car. A receiver on the vehicle will interpret these signals and control the motors through a motor driver, enabling real-time and smooth response to gestures.

For safety, the car will include a front-facing IR or ultrasonic sensor to detect obstacles and stop or slow down automatically, preventing collisions in case of errors or signal issues. The concept can also be expanded to a gesture-controlled drone with a camera for aerial movement and live video applications.

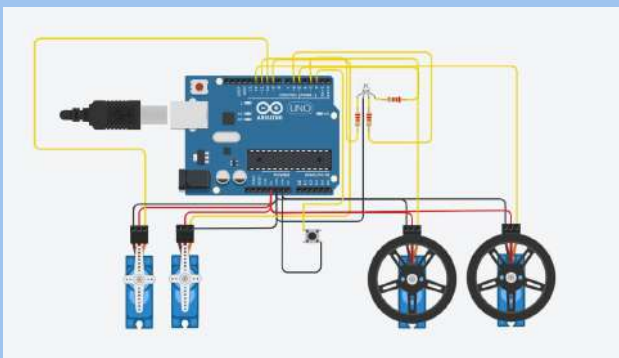


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18



Project Contributors

VARDAAN SAXENA

EVA - CUSTOMBUILT PERSONAL ROBOT

Eva is a custom-built personal robot designed to move, sense, express, and interact seamlessly in real time, creating a more engaging and lifelike human-robot interaction experience. At its core, Eva is powered by the reliable Arduino Uno R3, which coordinates multiple sensors, actuators, and output devices to deliver synchronized behavior and responsiveness. The robot integrates SG90 micro servo motors for expressive hand and head movements, allowing it to perform gestures that mimic human-like expressions, while an FS90R continuous servo enables smooth and controlled mobility.

Eva includes a tactile push button for easy manual control, while its face appears on a 0.96 inch OLED display showing animations and expressions for emotional interaction. An RGB LED system indicates moods through colors, and a DFPlayer Mini enables speech and sound. Battery-powered and portable, Eva combines visuals, controls, and audio with autonomous or semi-autonomous movement in a compact robotic companion.

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TECHNOGROW

TechnoGrow is an innovative, solar-powered hydroponic farming system designed to promote sustainable, efficient, and fully automated agriculture, particularly in environments where traditional farming faces limitations. By harnessing renewable solar energy through integrated panels, the system independently powers essential components such as water pumps, nutrient circulation, and monitoring units, reducing dependence on conventional electricity sources. Instead of soil, TechnoGrow uses a hydroponic setup where plants grow in a nutrient-rich water solution, allowing for faster growth, better resource utilization, and reduced water wastage.

The system uses sensors to monitor temperature, humidity, and methane levels, enabling real-time adjustments and safety alerts to maintain optimal plant conditions while reducing manual effort. TechnoGrow also features IoT-based remote monitoring with a custom dashboard built using Python, offering live data, analytics, and insights to support efficient and sustainable farming.

Its automated setup uses sensors and IoT monitoring to maintain optimal growing conditions while minimizing resource use and manual effort.

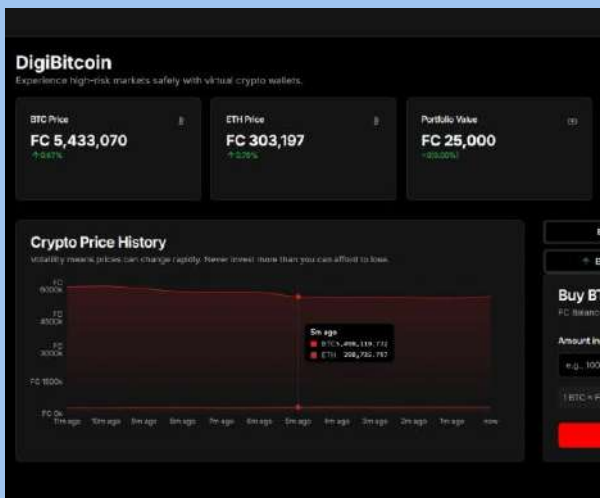


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20



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21

FINOVO - FINTECH MEETS INNOVATION FOR THE NEXT GENERATION

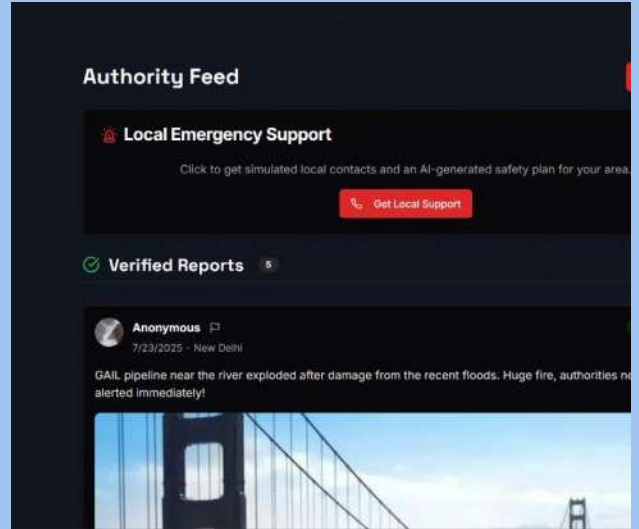
Finovo is a smart, secure, and educational fintech app designed to help kids learn money management in a fun and responsible way. It features Google/Microsoft login, an in-app Finovo Currency, a personalized dashboard with budgeting, balance, transactions, rewards, and QR payments, along with safe money transfer options and insightful spending reports. Finovo also includes learning simulators like ITR filing, digital investment practice (gold, bitcoin, stocks), a "Be a Banker" simulation, and a Scam Hunter game to teach online safety. With strong parental controls and activity monitoring, Finovo blends safety, learning, and innovation to build financially smart young users.

By combining practical tools, simulations, and gamified learning, it helps kids understand saving, spending, and digital transactions in a safe environment. With built-in parental oversight, it balances independence and guidance while encouraging responsible financial habits from an early age.

GROUNDTRUTH

GroundTruth™ is an AI-powered disaster rescue platform designed to keep communities informed, connected, and safer during natural disasters by turning real-time ground data into actionable insights. It offers Google-based login, role-based dashboards for users, admins, and authorities, multilingual support, and an interactive dashboard with themes, profiles, and an honor score system. Key features include AI chatbot assistance, nearby emergency contacts, multi-format reporting (text, video, voice), satellite map views, live report trends, verification status tracking, notifications, national news, emergency services directory, community chat, and rapid alert dispatch. GroundTruth™ aims to strengthen disaster response, speed up help, and build resilient, tech-enabled communities.

Eva is designed as an interactive robotic companion that blends simple controls with expressive feedback. Its compact form integrates visual, audio, and lighting elements to create a more engaging user experience. With portability and flexible operation modes, it serves as both a functional device and a creative platform for exploring human-robot interaction.



Project Contributors

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Saransh Wadhwa

Simran



22



Project Contributor

Kumar Shaurya



23

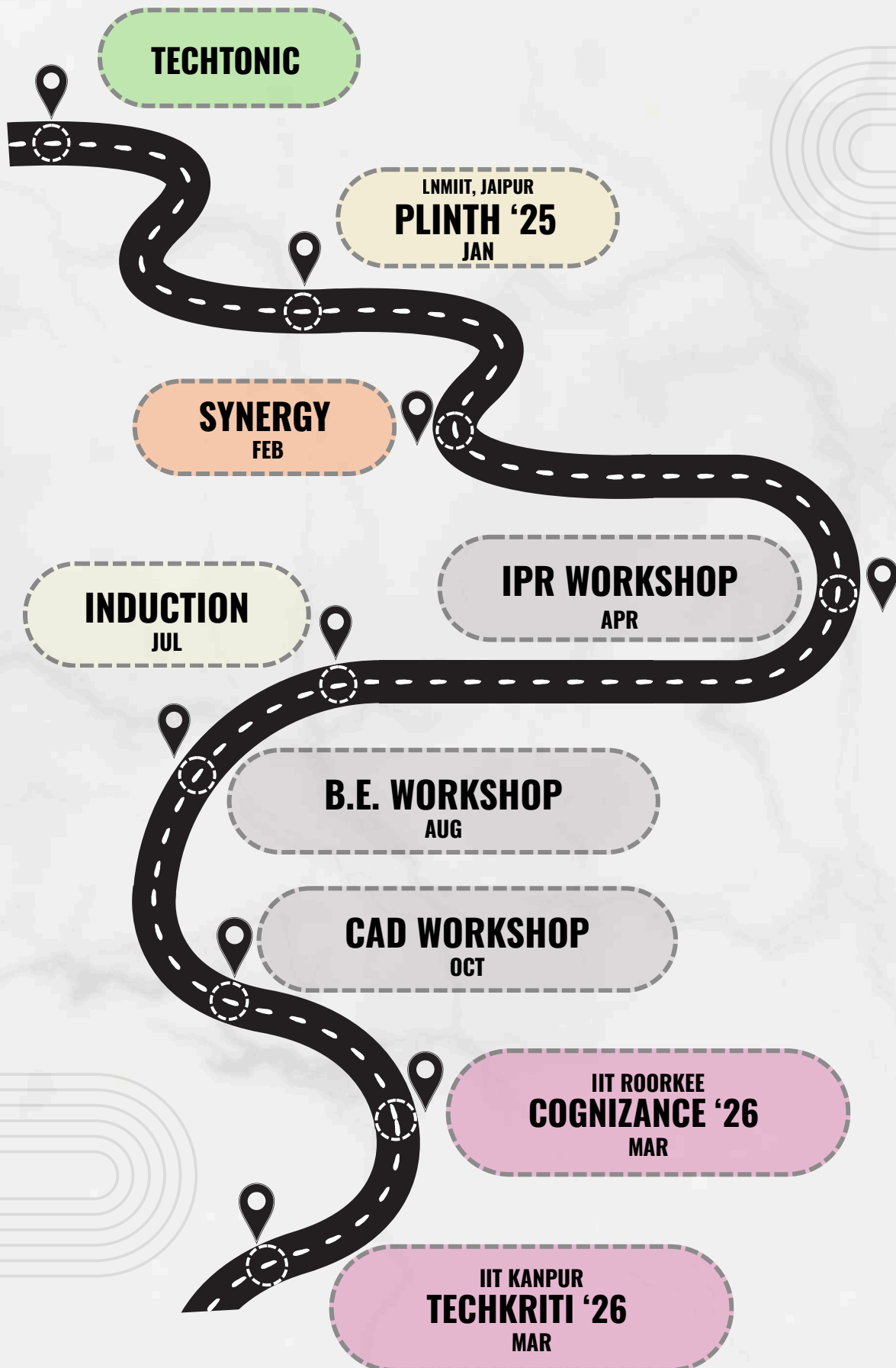
CICR X CONVERGE'26 WEBSITE

Converge'26 is the official techno-cultural fest platform developed for CICR, IIIT Noida, offering an immersive and engaging digital experience. Inspired by the Marvel & DC universe, the website blends creativity with functionality to capture the fest's high-energy spirit.

It serves as a centralized hub for event discovery, seamless registrations, and real-time updates, ensuring a smooth user experience. The platform features detailed information on flagship events like RoboRace and RoboSoccer, along with various technical and fun competitions.

With an interactive interface, dynamic visuals, and an integrated event gallery, the website highlights the legacy and vibrance of CICR events. Designed for scalability, it simplifies participation while enhancing accessibility and engagement, making it a powerful digital gateway for Converge'26.

ACTIVITIES



TECHTONIC 1.0

Annual Robotics Workshop



BASICS OF ROBOTICS

DAY 1

The workshop started off with an overview of some basics related to robotics, including important topics such as sensors, actuators, and microcontrollers. Hands-on experience was gained in constructing basic robotic systems.

PCB DESIGNING

DAY 2

The next day was devoted to Printed Circuit Board (PCB) designing, an important competency in today's robotics and electronics industries. The students were introduced to circuit diagramming, PCB designing techniques, and the fundamentals of manufacturing PCBs.



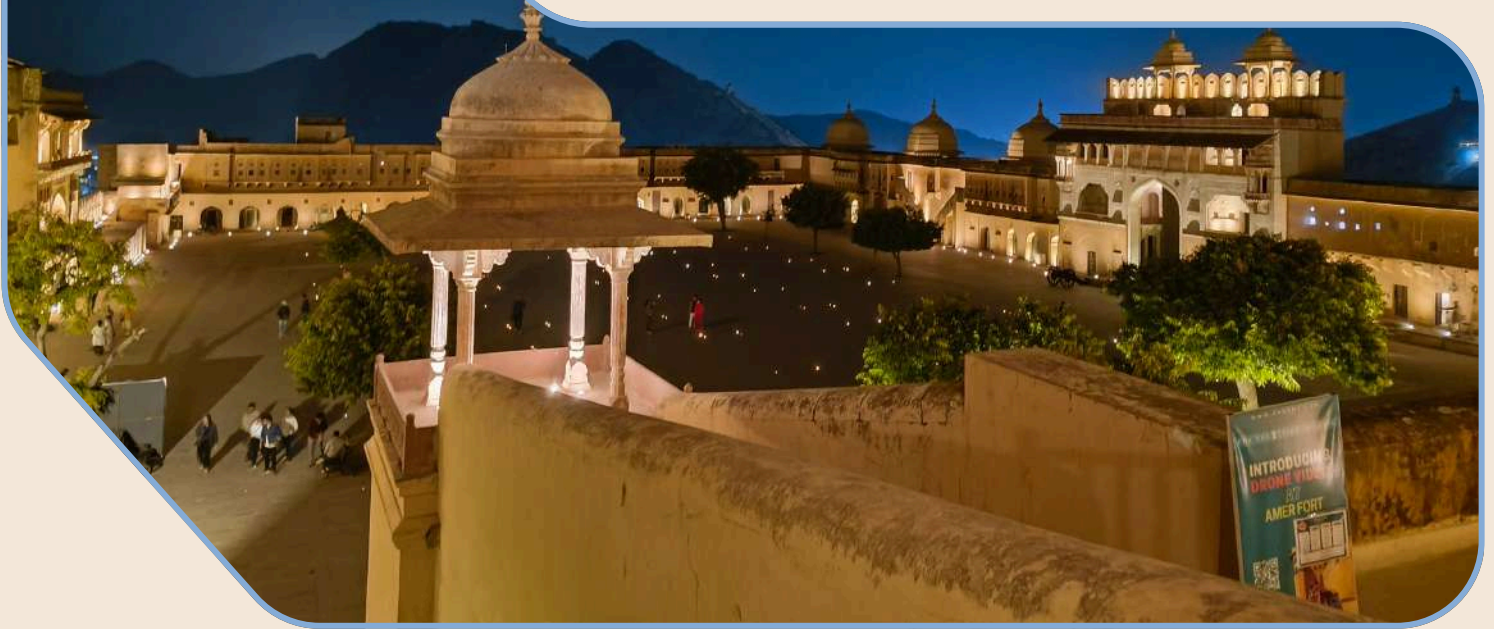
3D MODELLING

DAY 3

On the last day, learners were taught how to do 3D modeling and design, enabling them to think about and model robotic parts. Using CAD software that was relevant in the industry, students modeled different robot parts.



PLINTH '25



On 24th January, a team of CICR members embarked on a 5-day trip to Jaipur in a Traveller vehicle to participate in PLINTH, the annual tech fest of LNMIIT Jaipur. The journey began at 5:00 AM, filled with excitement and enthusiasm for both the competition as well as exploring.

Our team took part in various competitions like Robo Soccer and Robo Race, testing both our technical skills and decision-making abilities in the process.

EVENT PARTICIPATION

CICR members formed multiple teams and actively participated in various competitions held in PLINTH'25, LNMIIT-Jaipur, showcasing their technical expertise, teamwork, and problem-solving abilities.

EXPLORING JAIPUR

The trip extended beyond the competitions, offering the team a chance to experience Jaipur's rich heritage. Visits to iconic landmarks like the Hawa Mahal, Jantar Mantar, and various historical forts, which provided the team with cultural insights to complement their technical exposure.

The PLINTH trip was a perfect blend of learning, competition, networking, and exploration, leaving all the participants with unforgettable memories as well as valuable insights to bring back to CICR.

Jaipur

Apart from competitions, participants loved the vibrant city environment. A highlight of the fest was the Disco Night. It brought students from across the country together to celebrate and enjoy the festivities.



SYNERGY

CICR at Synergy

Innovation in Action

With planning, teamwork, and determination, the club brought exciting robotics-based events to the annual fest Synergy. The events showed real engineering skills and creativity along with the blend of fun. Through the variety of events conducted, CICR showed the impeccable spirit of innovation serving the club's true ideology, fostering creativity and practical engineering skills among its members.





EVENT HIGHLIGHTS

Robo Soccer

- Custom-built robots played real-time football matches
- Controlled live by participants during gameplay
- Every goal brought loud cheers from the crowd
- Showcased teamwork, strategy, and engineering skills



Fast-paced matches where robotics meets football.

Robo Race

- Robots raced through obstacle-filled tracks
- Included sharp turns, ramps, and barriers
- Required precision control and quick decisions
- Focused on speed, stability, and performance



A test of speed, control, and smart navigation.

CELEBRATING INNOVATION & AWARENESS

IPR WORKSHOP

AWARENESS SESSION & EXPERT INTERACTION



About the Workshop

CICR, in partnership with United & United, conducted an IPR Awareness Session on 26 April in celebration of World Intellectual Property Day.

The session emphasized the significance of Intellectual Property Rights in today's innovation-centric environment, particularly for students engaged in startups, projects, and technological concepts.



An insightful session with expert speakers and student participation, making the event highly engaging and impactful.



The session covered key aspects of patents, copyrights, trademarks, and design protection, helping participants understand how to legally safeguard their innovations. Through practical explanations and real-world examples, students gained clarity on applying IPR in academic and entrepreneurial journeys.

A highlight of the session was the live interaction with IP experts, where students received guidance on protecting their ideas and projects. The speakers, Mr. Vivek Srivastava (Patent Specialist) and Mr. Akash Chaudhary (IPR Expert), provided valuable insights and practical knowledge, making the session highly informative.

WELCOMING THE NEXT GENERATION

INDUCTION'25

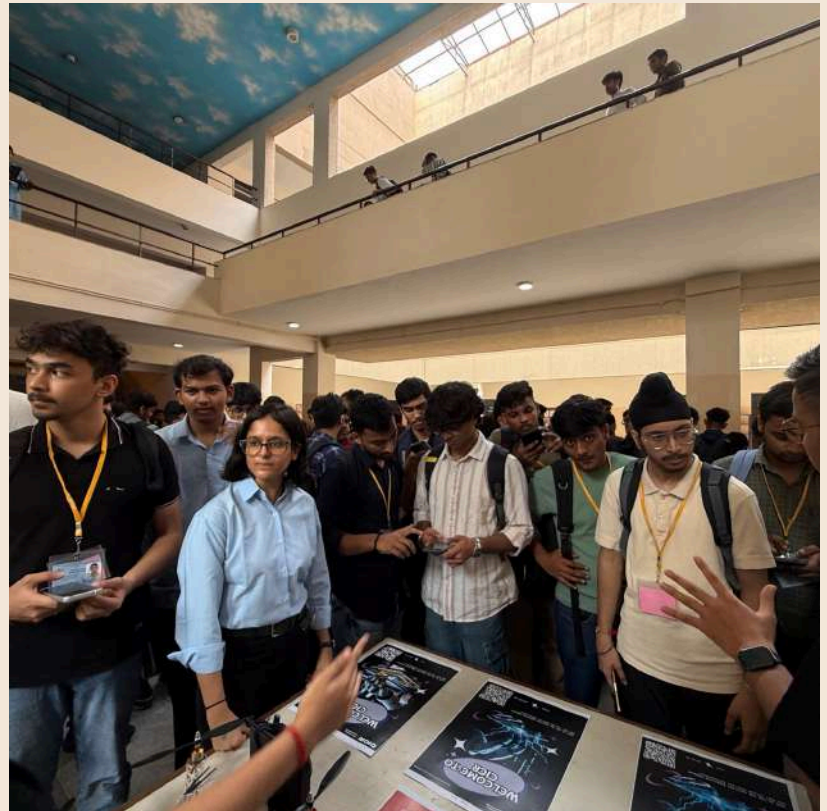
TECH SHOWCASE & MINI GAMES



The Tech Showcase

To give freshers a glimpse of what we build, A hands-on tech exhibition was hosted in the MPH. Students interacted with custom electronics games developed by the team, including:

- Path Finder & Buzz Wire
- Matrix Maze & Memory Game
- Snake on Matrix Display



A massive success with over 100 new enrollments, laying a strong foundation for an innovative and highly productive year ahead!

The CICR official induction served as the perfect launchpad for our new members, introducing them to our vision, ongoing projects, and the fast-evolving world of Robotics and AI. Our core team delivered an engaging orientation, outlining the club's technical roadmap and emphasizing our culture of collaborative, curiosity-driven learning. Following the presentation, seniors interacted directly with the juniors, clarifying doubts about coding, hardware, and how to get involved in upcoming hackathons and projects.

BUILDING FOUNDATIONS IN ELECTRONICS

B.E WORKSHOP

HANDS-ON LEARNING & PRACTICAL EXPERIENCE



About the Workshop

CICR held a Basic Electronics Workshop for 1st and 2nd year students exploring Arduino, microcontrollers, and circuit design. Students got hands on experience with components and explored display projects getting a real feel for how electronics and programming come together in the real world.



An interactive session with enthusiastic participation, encouraging creativity and real-world design thinking among students.

The workshop was interactive with students building and testing circuits through the session. 2nd year mentors guided them at every step making it easier to follow and understand how Arduino works and how different components come together in a system.

By the end students had a much clearer understanding of Arduino and basic electronics. Working hands on with seniors built their confidence and left many of them eager to learn more and build projects of their own.

BUILDING FOUNDATIONS IN DESIGN

CAD WORKSHOP

HANDS-ON LEARNING & PRACTICAL EXPERIENCE

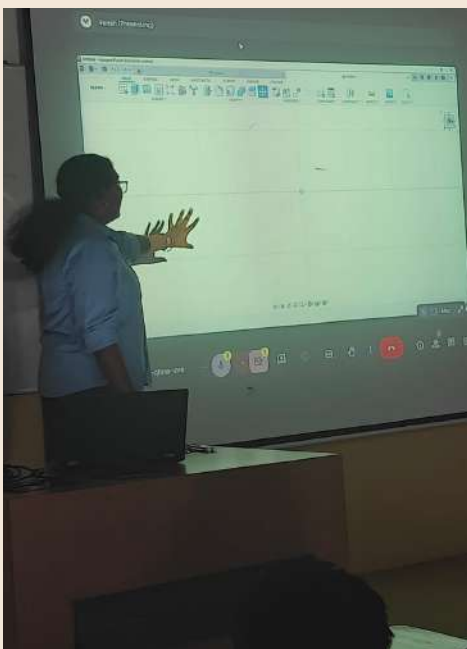


About the Workshop

CICR ran a CAD workshop for first year students in October 2025, introducing them to designing through Fusion 360. Rather than just talking through concepts, the session was kept interactive so students could follow along and get hands on with things like 2D sketching, 3D modeling, dimensioning and constraints from the very start.



An engaging session with enthusiastic participation, fostering curiosity and practical learning among students.



The workshop was centered around actually doing things rather than just listening. Students worked through guided exercises and somewhere along the way got to design their own keychain which honestly made it a lot more fun and gave them a chance to use what they had picked up in a creative way.

It was a good example of what CICR tries to do, keep learning practical, creative and useful. By the end of the session students felt a lot more comfortable using design tools, had a better grip on the basics and many of them left curious to explore more of what CAD has to offer.



Cognizance '26

NATIONAL LEVEL TECHNICAL FEST



CICR members came together in multiple teams and jumped into various robotics events at Cognizance 2026, IIT Roorkee. The competitions threw a lot at them, from movement and grip to autonomous navigation and overall performance. Teams had to design, build and keep improving their robots as things went on which came down to quick thinking, good teamwork and getting their hardware and software to work together when it mattered most.

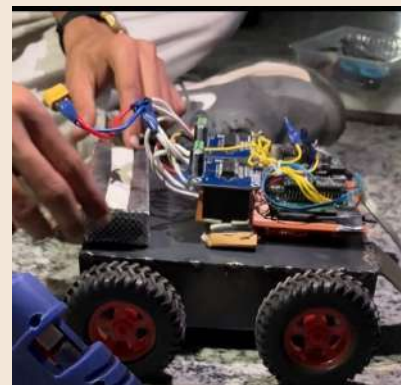
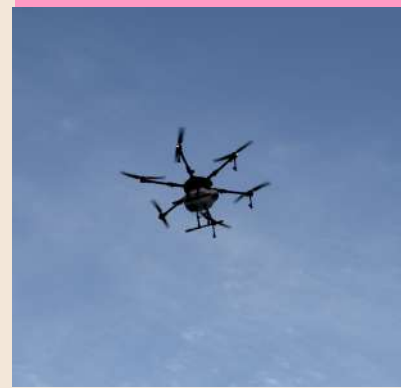
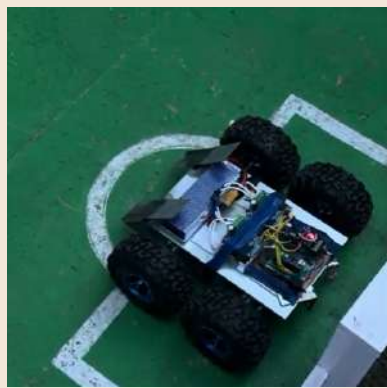
On 12th March a group of CICR members made their way to Roorkee for Cognizance, IIT Roorkee's annual technical fest. The trip had a great vibe to it as teams got ready to compete in different events. But it was more than just competing, it was a chance to explore the fest, see what others were building and get a real feel for what a national level technical environment looks like.

OVERVIEW

The Gallery

COMPETING AT COGNIZANCE WAS A CHANCE FOR CICR MEMBERS TO TAKE WHAT THEY HAD LEARNED IN THE CLASSROOM AND ACTUALLY USE IT. WORKING UNDER TIME PRESSURE PUSHED THE TEAM TO DESIGN, BUILD AND FIX THINGS ON THE GO WHILE ADAPTING AS CHALLENGES CAME UP. FROM TESTING ROBOTS TO SOLVING PROBLEMS IN THE MOMENT IT ALL CAME TOGETHER TO BUILD STRONGER TEAMWORK, SHARPER THINKING AND THE KIND OF CONFIDENCE THAT ONLY COMES FROM DOING THINGS IN A REAL COMPETITIVE SETTING.

IIT ROORKEE



Acheivements

CICR MEMBERS DID REALLY WELL AT SOME WELL KNOWN COMPETITIONS, WITH TEAMS FINISHING 3RD IN BOTH THE CASE STUDY COMPETITION AND THE AI MODEL TRAINING COMPETITION. IT WAS A GREAT SHOWING FROM THE PARTICIPANTS WHO BROUGHT SOLID THINKING, TECHNICAL SKILLS AND GOOD TEAMWORK TO THE TABLE, AND A PROUD MOMENT FOR THE CLUB AS A WHOLE.

IIT
ROORKEE



Case Study Competition

The Team **Frameflux** consisted of two 1st-year students.

- Shreya Bhatt
- Akansha Nagar



AI Model Training Competition

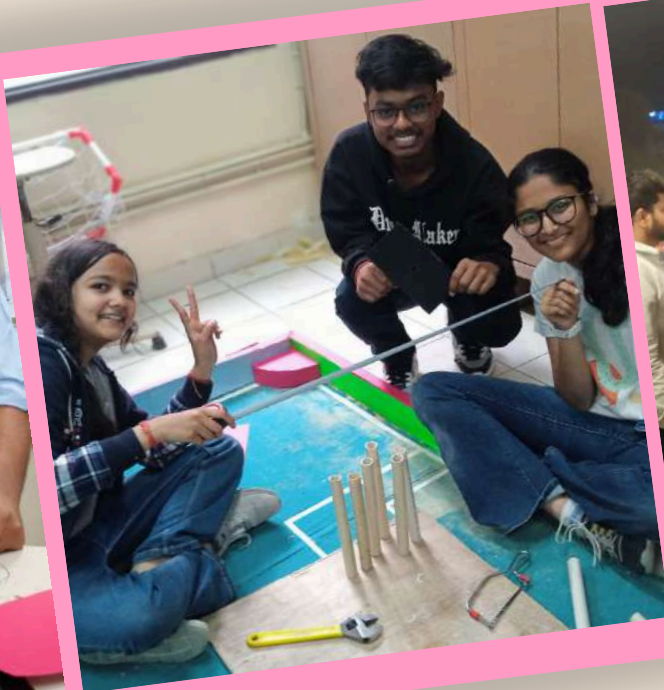
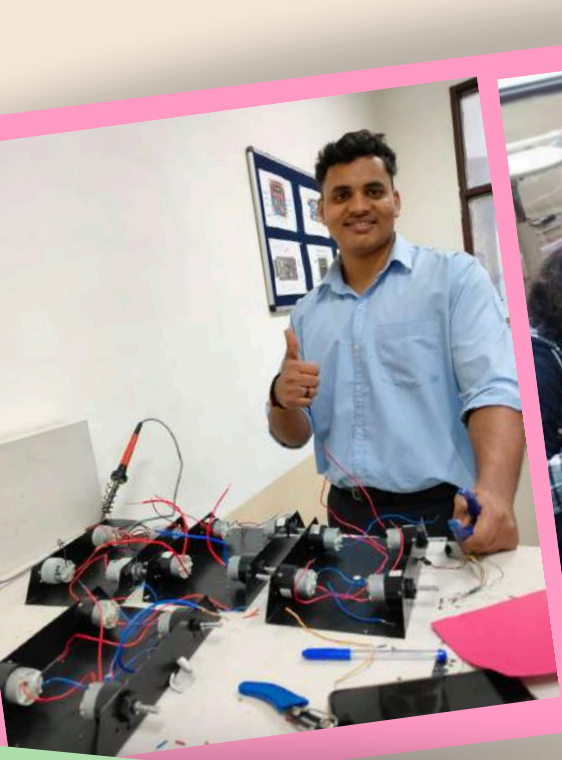
The Team **Bellatrix** consisted of five 2nd-year students, in which three consisted of CICR Students

- Ankit Sharma (CICR)
- Shreya Singh (CICR)
- Shriya Singh (CICR)
- Sarthak siddhant
- Ujjwal Parashar

GALLERY



GALLERY



VISION FOR 2026-27

As we step into the new academic year, CICR remains committed to curiosity, creation, and collaboration. Our vision for 2026–27 is simple: strengthen the culture that makes this club unique while pushing ourselves to achieve more, together.



Step By Step We Make It

01 Preserving Our Core Culture

Strengthen the culture of innovation, teamwork, and continuous learning.

02 Strengthening Alumni Bonds

Reconnect alumni with students through mentorship and collaboration.

03 Scaling Achievements

Scale technical achievements through larger projects and competitions.

04 Expanding Impact Beyond

Expand CICR's impact across campuses, communities, and industries.





The Team of Resilience & Growth

THANK YOU

We appreciate your support
and encouragement

Design Lead :
Ashutosh Vishwakarma

This journey has been more than just building projects
—it has been about learning, failing, growing, and
creating together.

Every idea, every late-night discussion, and every
small breakthrough has shaped who we are as a team.
CICR is not just about technology; it's about the people
behind it—the passion, the effort, and the shared vision
to build something meaningful.

We are grateful for every bit of support that helped us
reach here, and we're just getting started.

Scan to visit
www.cicr.in

